TINKERBADGE / ELECTRICAL OVERVIEW

Prototype schematic 0.1 | 2026-11-09
Source-controlled circuit contracts: docs/design/
Electrical source review precedes each populated block.

01 POWER + CHARGING

USB / dock input selection
Power path, protected BAT sensing
Single nominal 3.3 V shared rail

File: power.kicad_sch

02 PROCESSOR + RECOVERY

ESP32-S3-WROOM-1-N16R8
Native USB, UART and boot/reset
Reviewed GPIO allocation

File: mcu.kicad_sch

03 E-PAPER DISPLAY

ED037TC1 electrical interface
Switched logic and isolated inputs
Exact panel boost support

File: display.kicad_sch

04 LIGHTING + BUTTONS

Three role + five affinity pixels
Red / green power indicator
Two buttons, RTC wake

File: lighting.kicad_sch

05 SENSORS + EXPANSION

Ambient, motion, dynamic NFC
IR transmit / receive
Qwiic / SAO and status expander

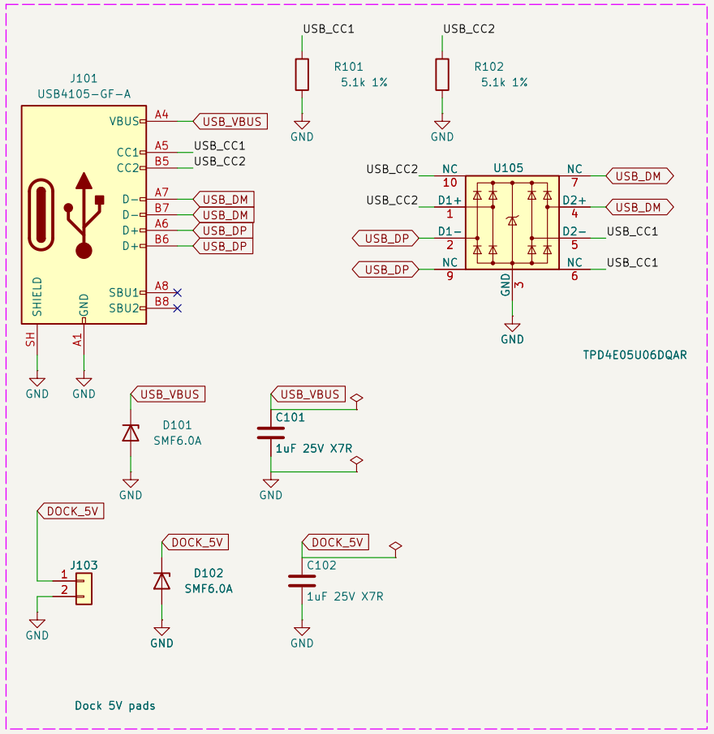
File: peripherals.kicad_sch

INITIAL VARIANT
Onboard charging is disabled.
Two-wire protected LiPo; optional NTC not fitted.
See docs/design/implementation_status.md
for physical and prototype release gates.

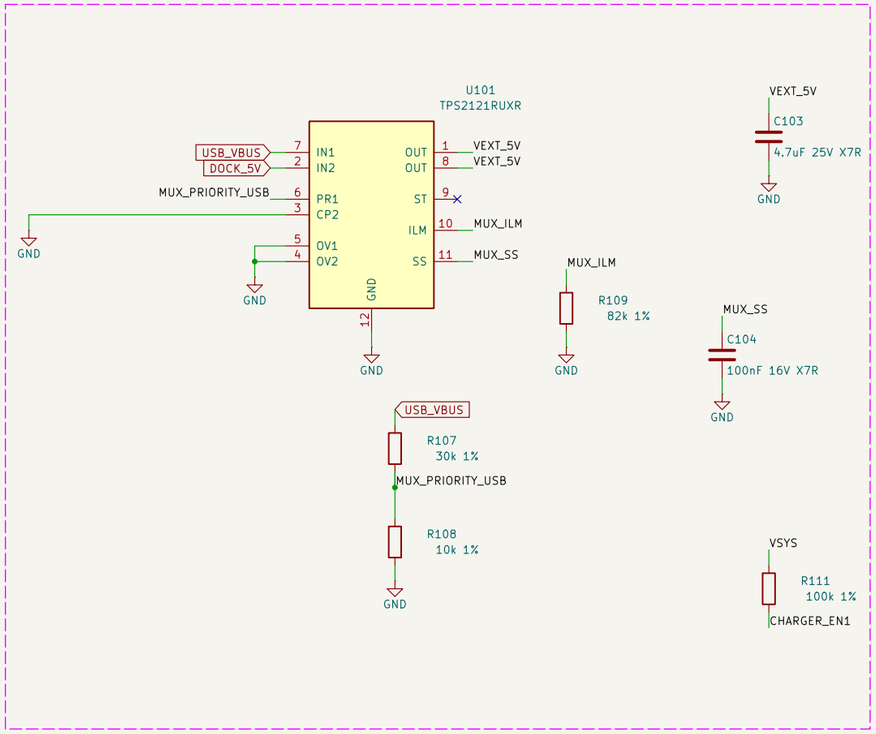
PROTOTYPE RAIL DECISION
Shared 3.3 V supply, including the eight RGB LEDs.
Tolerance can fall below the LED specified minimum.
Startup / low-battery / Wi-Fi / LED-step tests required.

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File: tinkerbadge.kicad_sch		
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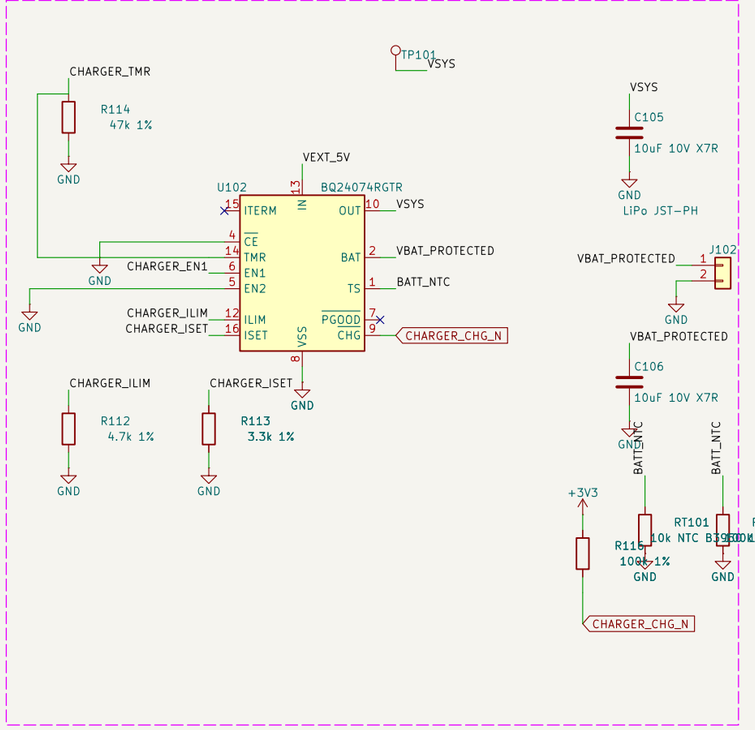
USB / dock inputs



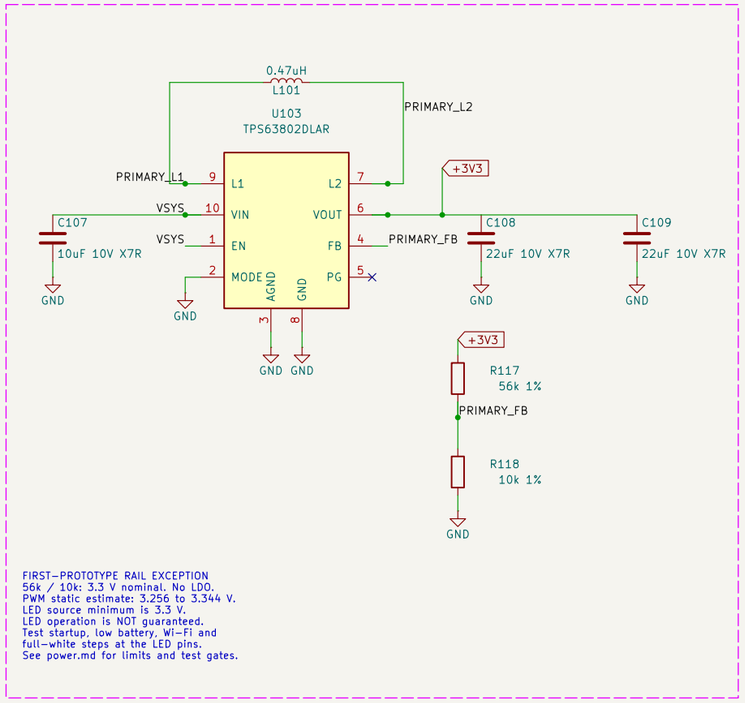
USB-priority source selection



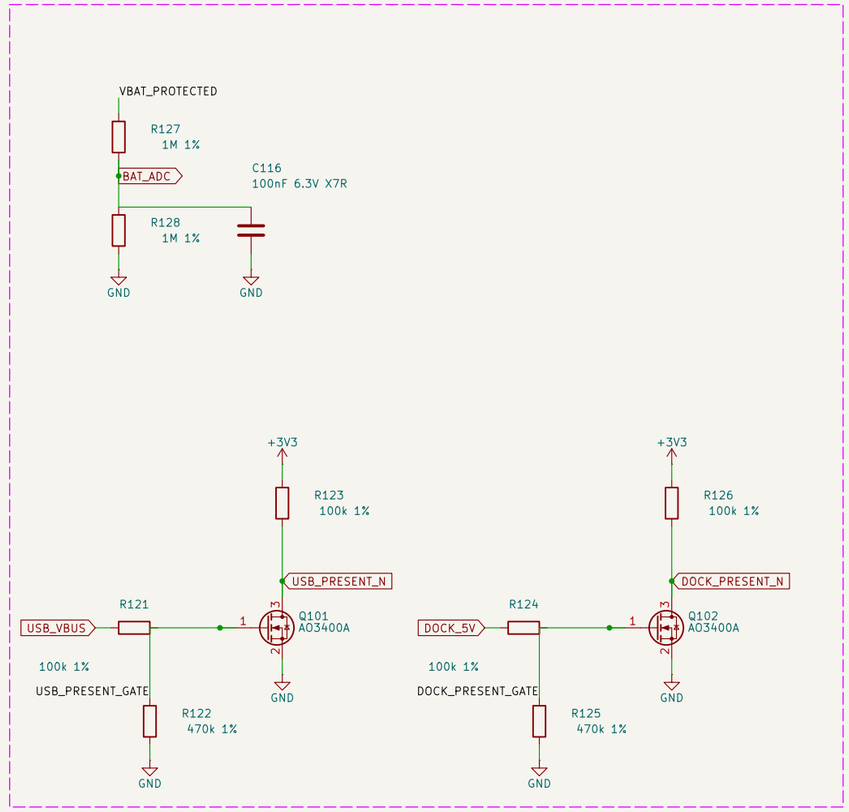
Power path: charger enabled



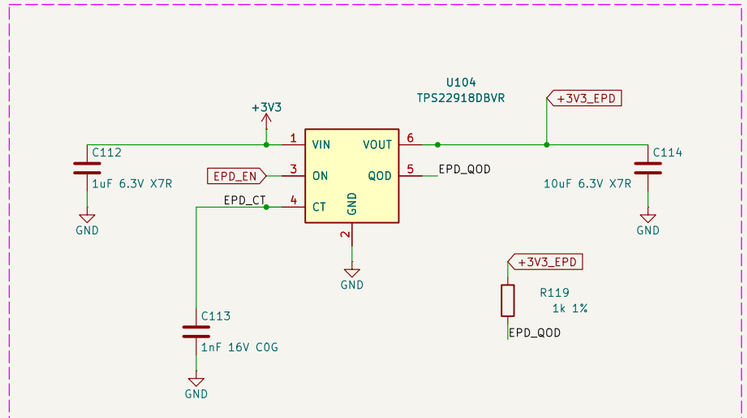
Shared nominal 3.3 V rail



Battery ADC (VBAT/2) / source presence



Display power



CHARGING
CE tied low: charger enabled, 270 mA (R113), USB500 input limit.
RT101 10k || RT102 100k NTC under the cell: charge window 1.7-44.8 degC nominal,
0.1-48.4 degC over the charger's threshold and bias tolerance.
J102 is JST-PH, pin 1 = battery +. CHECK PACK POLARITY before connecting.
Use a protected 1S 4.2 V pack rated for >= 300 mA charge.

Electrical development draft. No PCB layout or manufacturing release. See docs/design/power.md.

Sheet: /01 POWER + CHARGING/		
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02 PROCESSOR + RECOVERY

ESP32-S3 Processor + Interfaces

+3V3
C201 22uF 10V X5R
C202 100nF 16V X7R
GND
Place 100 nF at VDD; keep bulk nearby.

U201
ESP32-S3-WROOM-1-N16R8

MCU_EN	3	EN			
		3V3	2		
		TXD0	37	UART_TX	
		RXD0	36	UART_RX	
BTN_B_N	27	IO0			
BAT_ADC	39	IO1			
BTN_A_N	38	IO2			
	15	IO3			
	4	IO4			
USB_PRESENT_N	5	IO5			
DOCK_PRESENT_N	6	IO6			
MOTION_INT1	7	IO7			
I2C_SDA	12	IO8			
I2C_SCL	17	IO9			
EPD_CS_N	18	IO10			
EPD_MOSI	19	IO11			
EPD_SCK	20	IO12			
EPD_DC	21	IO13			
EPD_RST_N	22	IO14			
EPD_BUSY	8	IO15			
EXT_SDA	9	IO16			
		IO17	10	IR_TX	
		IO18	11	IR_RX_N	
		USB_D-	13	MCU_USB_DM	
		USB_D+	14	MCU_USB_DP	
		IO21	23	RGB_DATA	
		PSRAM IO35	28		
		PSRAM IO36	29		
		PSRAM IO37	30		
		IO38	31	PWR_LED_G	
		IO39	32	PWR_LED_R	
		IO40	33	EPD_EN	
		IO41	34	EXT_SCL	
		IO42	35	SAO_GPIO1	
		IO45	26	STRAP45	
		IO46	16	STRAP46	
		IO47	24	SAO_GPIO2	
		IO48	25	NFC_FD_N	
		GND	1	GND	

UART RECOVERY

TP205 UART_TX
TP206 UART_RX

USB 2.0 FS

R205 22R 1%
R206 22R 1%
MCU_USB_DM → USB_DM
MCU_USB_DP → USB_DP

BOOT STRAPS

STRAP45 — R203 10k 1% — GND
STRAP46 — R204 10k 1% — GND
Default: SPI boot, 3.3 V flash.

DISPLAY DEFAULT OFF

EPD_EN — R207 100k 1% — GND

GPIO40 may go high in USB-OTG ROM recovery.
The panel remains reset by its local pull-down.

Enable + Reset

+3V3
R201 10k 1%
MCU_EN
TP201
C203 1uF 10V X7R
GND
EN: 10 k / 1 uF. SW403 resets.

Boot / Download

+3V3
R202 10k 1%
TP202
BTN_B_N
Button B is BOOT: hold B, tap RESET.

Power Test

+3V3 TP203
TP204 GND

RESET (recessed): SW403 pulls EN low through 330R.

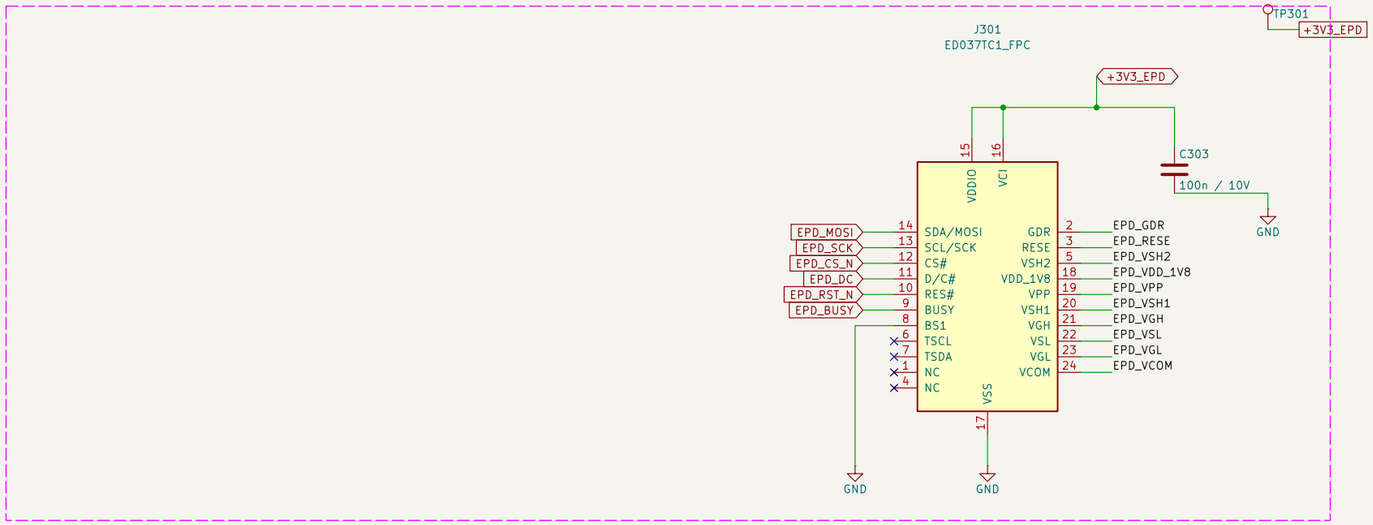
R415 330
SW403 SKSCLBE010
MCU_EN_SW
MCU_EN
GND

N16R8: GPIO35 / 36 / 37 reserved for octal PSRAM.
GPIO3 left open with default eFuse JTAG selection.
GND pins 1, 40 and exposed pad 41 are one net.
Antenna keepout and RF layout are PCB release gates.

Sheet: /02 PROCESSOR + RECOVERY/
File: mcu.kicad_sch
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Prototype electrical capture. Physical FPC mating and panel revision must be verified before layout.

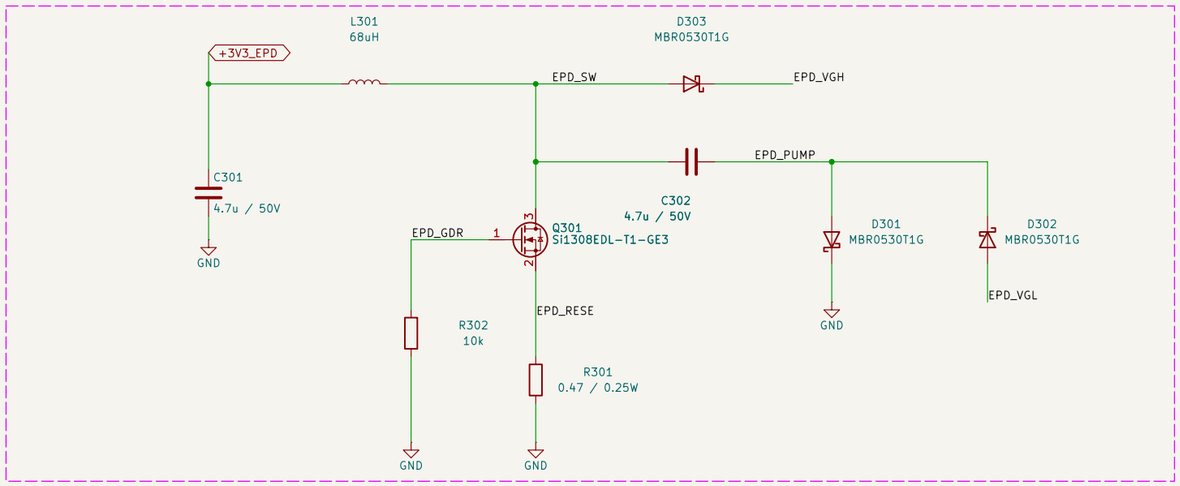
Panel interface / 3.7-inch display



MCU drives the panel directly. Firmware must drive SPI/DC/RST low before EPD_EN goes low.

Power-on: hold reset low, enable +3V3_EPD, initialize, then update. Quiesce SPI and sleep the panel before power-off.

Panel boost and charge pump

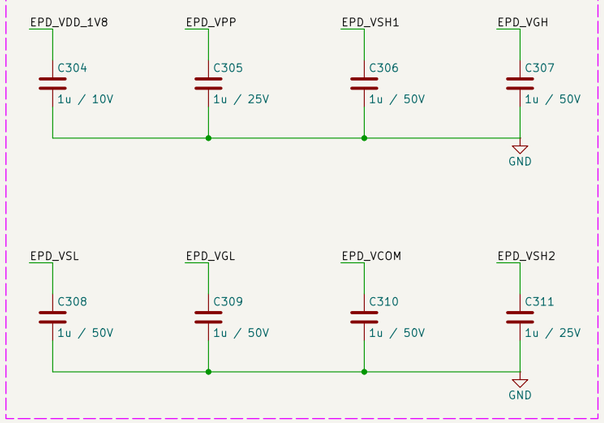


BUSY bias



Panel rail decoupling

Use nonpolar X7R capacitors; verify capacitance under DC bias.



FPC pins 1 and 4 are NC per the raw-panel document.

Pin 5 is VSH2. VDD on pin 18 is local 1.8 V, not the 3.3 V input.

Release gates: exact panel, FPC orientation, inductor saturation,

capacitor derating, measured startup/refresh/off-state waveforms.

J301: FH34SRJ top-and-bottom contact. Pads are numbered in panel order for the folded flex; dry-fit a real panel.

Sheet: /03 E-PAPER DISPLAY/		
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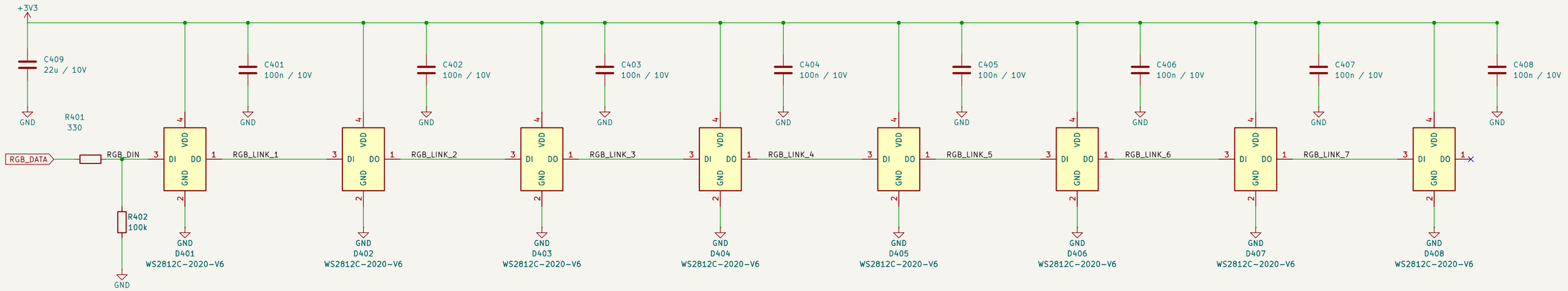
04 LIGHTING + BUTTONS

Addressable RGB Chain

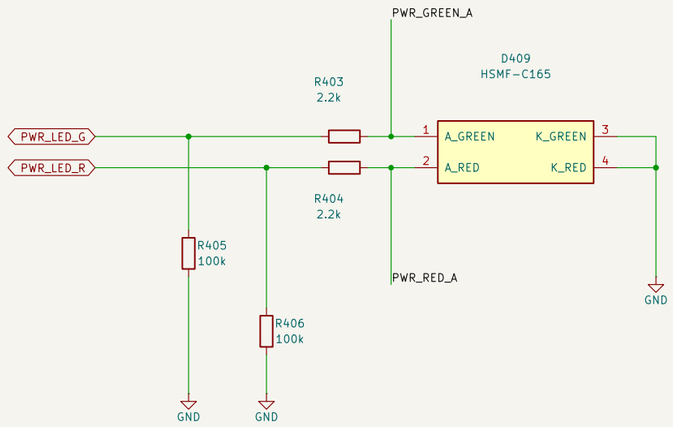
RGB chain: exact WS2812C-2020-V6 on shared nominal +3V3. Below 3.3 V operation is not guaranteed.

Send black before storage. Respect the system power/current budget in firmware.

Prototype gate: scope startup, low battery, Wi-Fi and full-white transitions at the LEDs; verify shared peak capacity.



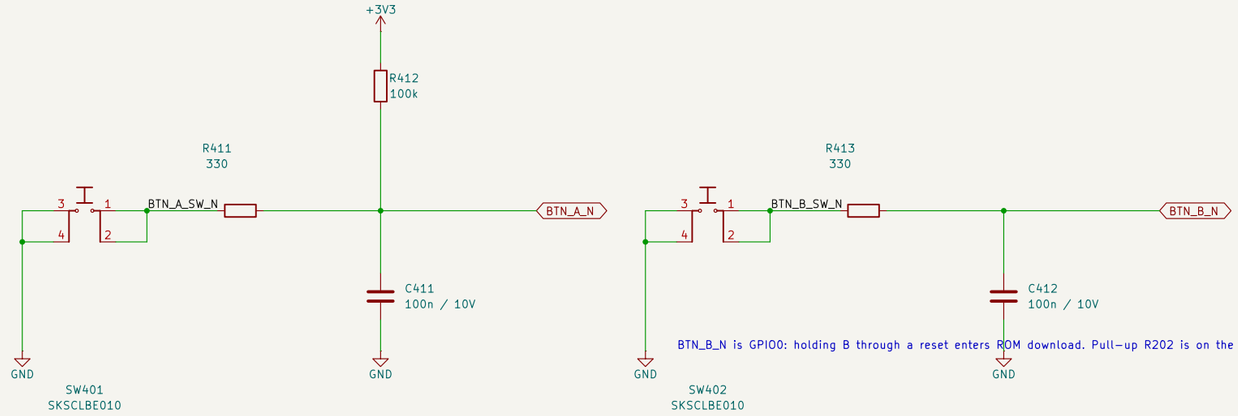
Independent Green / Red Power Indicator



Both LED control outputs default low. Current is limited by 2.2k per die.

User Buttons

A is the retained storage-wake button; B is a normal input.



BTN_B_N is GPIO0: holding B through a reset enters ROM download. Pull-up R202 is on the processor sheet.

Alps side push, 2.2 N. Common pairs: 1-2 and 3-4.

Always-on bias and capacitive loading need application qualification.

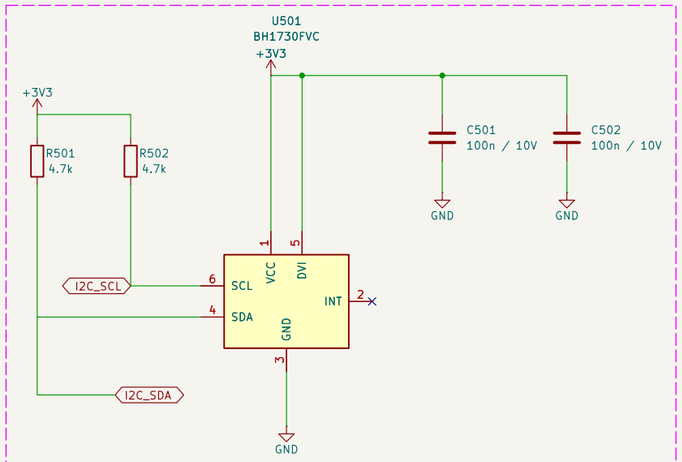
100k / 100nF gives about 10 ms debounce; firmware must also debounce.

Footprints for the exact LEDs and switch are intentionally blank pending their physical pad audit.

Sheet: /04 LIGHTING + BUTTONS/		
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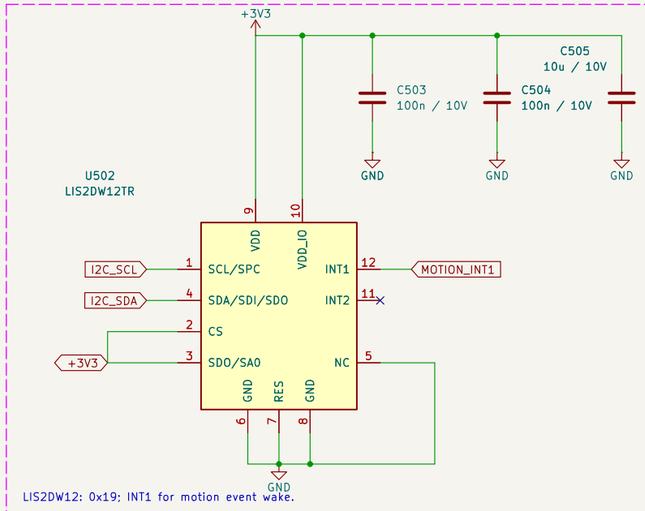
05 SENSORS + EXPANSION

Ambient Light + I2C Pull-ups



BH1730: 0x29; poll ambient light. INT unused.

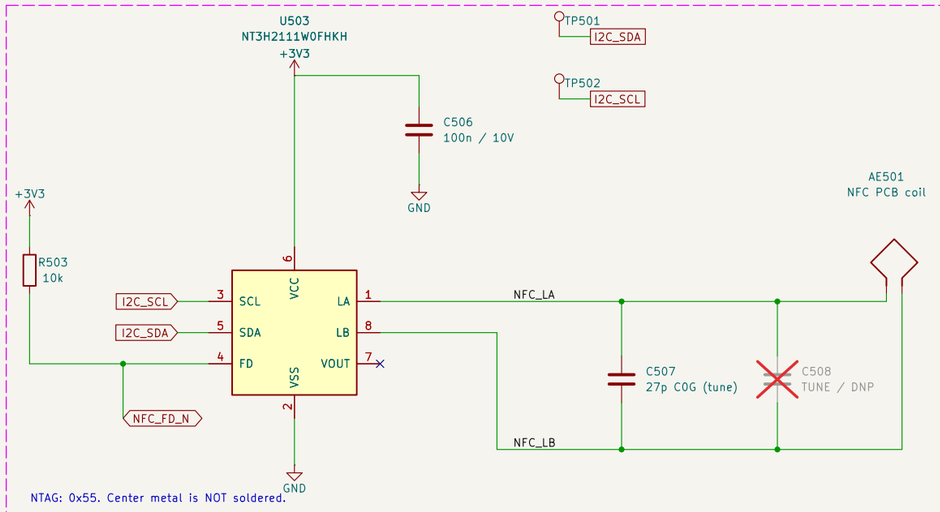
Motion Sensor



LIS2DW12: 0x19; INT1 for motion event wake.

CS and SA0 high. Reserved pad 7 is grounded.

Dynamic NFC Tag + Antenna

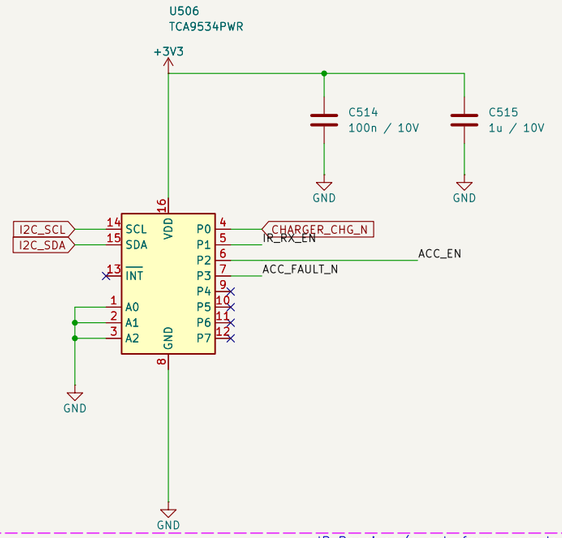


NTAG: 0x55. Center metal is NOT soldered.

Tune the PCB loop in the finished badge; tuning caps DNP.

ONBOARD I2C: 100 kHz initially. NTAG errata: STOP/START >= 50 us; guard RF edges; quiet bus for 1 ms at startup.

Status / Enable Expander



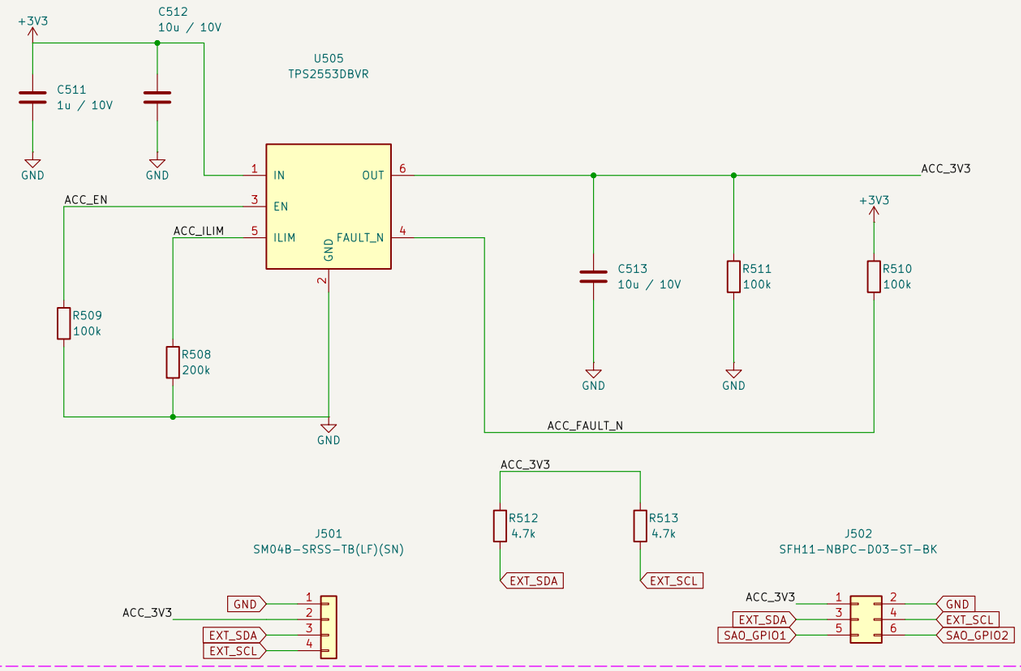
TCA9534 address: 0x20.
Write latch 0x00 before
direction register 0x09.

Protected Accessory Rail + Qwiic / SAO

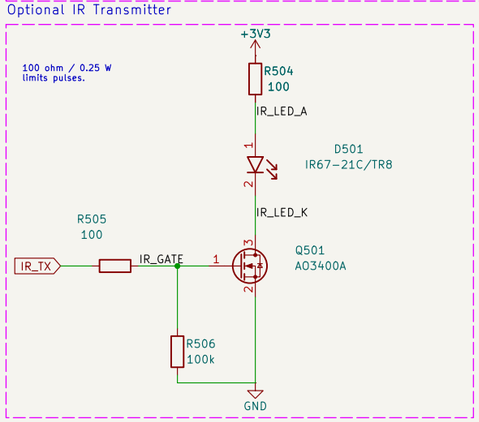
50 mA normal total. 200k / 1% sets roughly 115–160 mA current limit.

Qwiic + SAO: 3.3 V only, 50 mA shared budget.

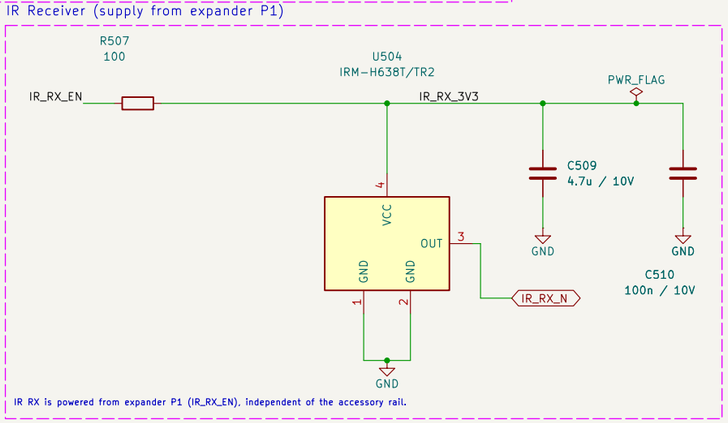
No hot-plug, external power or off-state GPIO drive.



Optional IR Transmitter



IR Receiver (supply from expander P1)



IR RX is powered from expander P1 (IR_RX_EN), independent of the accessory rail.

Sheet: /05 SENSORS + EXPANSION/
File: peripherals.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

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